

Emmanuel Idoro

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EDUCATION

Bachelor of Science, Software Engineering

University of Calgary, Schulich School of Engineering

09/2022 – 06/2028 | Calgary, Canada

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java), Software Design (C++), Databases, Operating Systems, Probability & Statistics.

SKILLS

Programming languages: Python, Java, SQL, C, C++, JavaScript, TypeScript, C#, HTML, CSS

Frameworks & Libraries: React, Django, FastAPI, Flask, REST APIs, scikit-learn

Tools/Technologies: AWS, Snowflake, Azure, Docker, Docker Compose, dbt, DuckDB, Git, GitHub, GitHub Actions, Prometheus, Grafana, Trivy, WebSockets, Figma, Postman, Microsoft Excel, Microsoft Teams

PROFESSIONAL EXPERIENCE

Royal Bank of Canada (RBC)

Data Engineer Intern

05/2026 – 12/2026 | Calgary, AB

- Validate internal and external **Snowflake data shares using SQL**, checking published views and schema mappings. Resolved a refresh issue that had left one share serving **stale data for four months**.
- Automate **API quota management** with a script that checks usage hourly and automatically increases quotas as usage approaches the limit to prevent quota-related service interruptions.
- Rotate **service and application secrets** and verify permissions for internal data shares, maintaining secure authentication and controlled access to enterprise data.

Mindrift AI

AI Model Trainer

03/2025 – Present | Remote

- Write **50** challenging Java/OOP prompts weekly (classes, inheritance, interfaces, exceptions) to stress-test AI model capabilities.
- Edit model answers into correct, runnable Java with brief explanations and unit tests, improving eval accuracy **~ 20%** on similar tasks.
- Review **10–20** contributor tasks per week for guideline compliance (score 1–5), reducing resubmits by **~25–35%** with targeted feedback.

Outlier AI

AI Model Trainer

09/2024 – 08/2025 | Remote

- Created training sets for JavaScript/React tasks that raised code-suggestion accuracy by **20%** on internal tests.
- Partnered with Quality Managers to tighten labeling rules and edge cases, cutting rework by **30%**.
- Analyzed evaluation mistakes and refined data/prompts, increasing solved tasks by **10%**.

PERSONAL PROJECTS

FinSight – Anomaly-Detection Web API (Python, FastAPI, DuckDB, dbt, scikit-learn, React, Docker)

- Built an end-to-end anomaly-detection pipeline that ingests **50K** synthetic banking transactions, transforms them through a dbt dimensional model (staging → intermediate → marts), and scores them with an Isolation Forest via FastAPI.
- Modeled a DuckDB star schema (**fact + 3 dimension tables**) with dbt-managed transformations and data quality tests, powering a React dashboard with anomaly explanations ranked by severity.
- Containerized the API and frontend into a Docker Compose stack with a single-command startup that runs data generation, dbt transforms, model training, and scoring automatically on first boot.

PipelineX – DevOps / SRE Layer for FinSight (GitHub Actions, Docker Compose, Prometheus, Grafana, Trivy)

- Automated a **4-stage** CI/CD pipeline (lint → tests → build → Trivy scan) that blocks merges on quality or security failures.
- Configured production-style Docker Compose deployment with health checks across **4 services**, cutting manual restart time from minutes to seconds.
- Provisioned Prometheus (**15+ metrics**) and Grafana dashboards for latency, error rates, and restarts, enabling **sub-5-minute** incident diagnosis.

LabelForge – Multi-User Annotation Platform (React, TypeScript, Django REST Framework, PostgreSQL, JWT Auth, Docker)

- Developed a full-stack annotation platform with **3 role-based** workflows (Annotator, Reviewer, Admin) and a task state machine enforcing valid transitions at the API level (unclaimed → in-progress → submitted → approved/rejected).
- Created an innovative annotation tool with customizable label buttons and keyboard shortcuts, enhancing usability for **15+** reviewers and leading to improved task accuracy and a decrease in revision requests by **30%**.
- Shipped a quality dashboard with per-annotator metrics (rejection rate, avg time, throughput), daily approval charts, and label-distribution breakdowns, filterable by project.